Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep coarse positions and discarded detail distinct.

Use exact windows,negative maxima and complete boundaries;separate map changes from predictions of a specified head.

Keep one habit: Assess independent sums,maxima,means and the absence of implicit ReLU using checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Compute max and **mean pooling** under an explicit window rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Only the start at row 0 column 0 fits. The **output** shape is 1 by 1 under the complete-window rule.

Opening retrieval

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an **input** through a hidden representation and a decision rule

Reference example, not the only acceptable wording: Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a weight penalty from **prediction** error

Reference example, not the only acceptable wording: Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

Pictured end goal

By the end, I can explain pooling and **spatial information** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Convolution as a local operation

Learned representations

Regularisation and early stopping

For the compact alternative schedule, use two 45-minute sessions. Prepare a labelled 4 by 4 grid with four window outlines. State complete windows, stride 2 and no padding before any arithmetic.

Keep pooling separate from ReLU, and discuss loss of location through explicit collisions.

Reference for the floor-based pooling convention: https://docs.pytorch.org/docs/stable/generated/torch.nn.modules.pooling.MaxPool2d.html . No software library is required.

Outcomes

Compute max and **mean pooling** under an explicit window rule

Use collisions to explain lost **spatial information**

Test boundary and movement claims with revealing examples

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach local summaries with exact spatial and boundary limits

> Distinguish same-window rearrangement from movement across a pooling boundary.

Main contract,complete arithmetic and guided key

A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

Max pooling keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

This code assumes a nonempty rectangular numeric map at least 2 by 2. It uses only complete 2 by 2 windows, stride 2 and no padding.

It returns two separate summaries of the same map; it does not apply one pooling operation after the other.

For this complete-window rule the **output** shape is floor((H - 2) / 2) + 1 by floor((W - 2) / 2) + 1. A 3 by 3 map produces just one **output**; its last row and last column are not read.

Do not silently invent a partial-window rule or pad missing positions.

Window start	Values in row-major order	Sum /4	Maximum
(0, 0)	[1, 4, 3, 2]	10/4=2.5	4
(0, 2)	[2, 0, 5, 1]	8/4=2	5
(2, 0)	[0, 6, 2, 4]	12/4=3	6
(2, 2)	[1, 3, 2, 7]	13/4=3.25	7

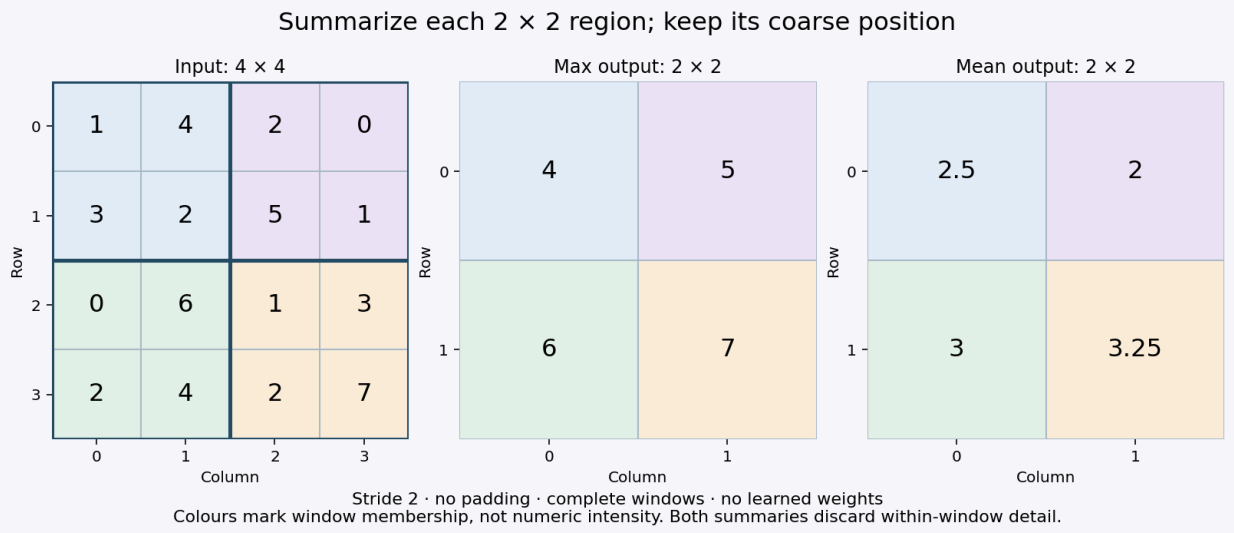
The two output maps preserve which coarse region produced each scalar. They do not preserve the four original positions and values within that region. pool 2 returns maxima and means separately from the original map. It requires a nonempty rectangular numeric input and checks that at least one 2×2 window fits. Only complete windows at stride 2 are read; with a3×3 input, the last row and column are excluded.

This boundary rule is part of the operation, not a discovered property of the data.

Guided: both windows have maximum 6 and mean 1.5. Either scalar alone cannot distinguish their different nonzero positions.

Exit: the first two visual windows collide, so exact reconstruction is impossible from the pooled values alone. The 8 crossing from column 1 to column 2 changes the pooled map, so universal shift invariance does not follow.

Four-region graphic and declared movement head



Complete2×4 movement probe, all other values 0	Max map	Mean map	Declared probe head:class 1if left max>0
8 atrow0, column 1	[[8, 0]]	[[2, 0]]	1
Move8 torow0, column 2 andzeroitsoldposition	[[0, 8]]	[[0, 2]]	0

Complete2×4 movement probe, all other values 0	Max map	Mean map	Declared probe head:class 1if left max>0
What follows	Crossing the window boundary changes both maps	Not universal one-pixel invariance	This explicitly supplied head changes class; unknown other heads need their own checks

Complete separate two-window picture

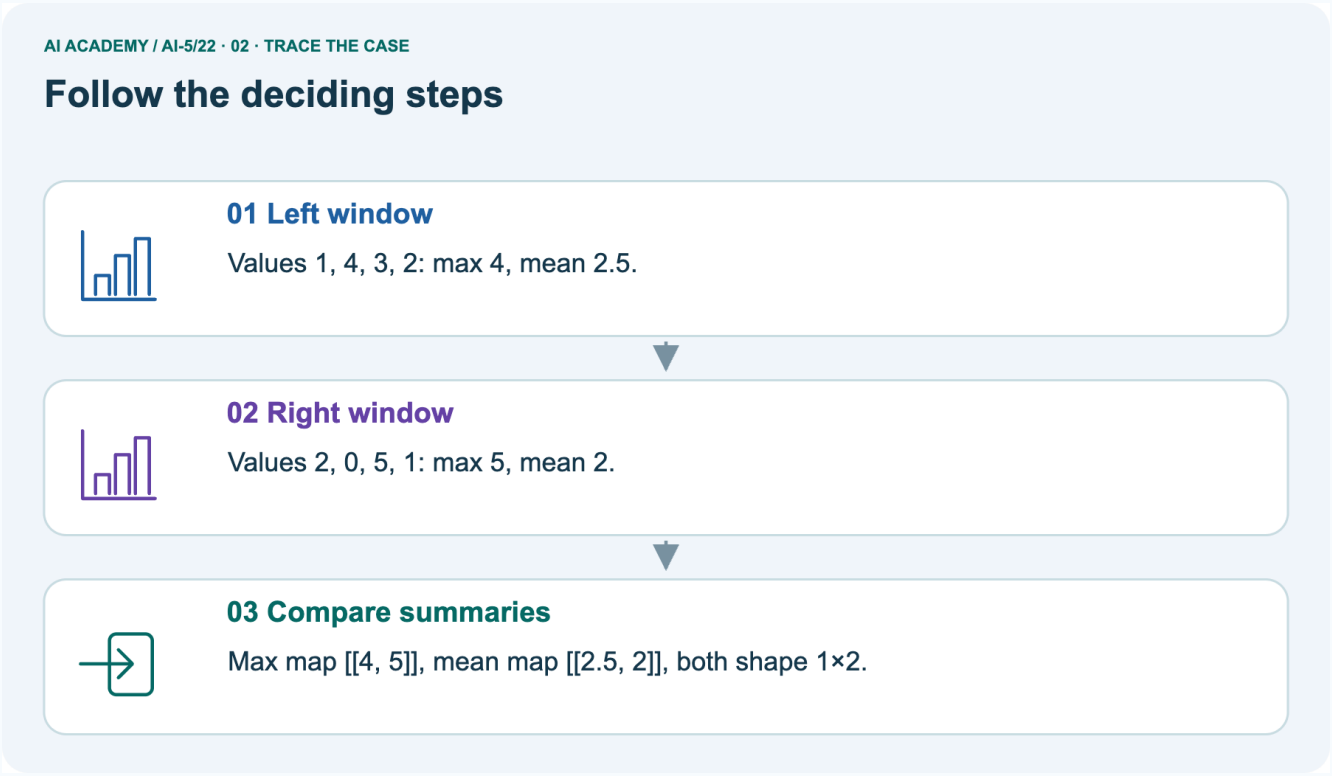
Pool a small map and show the detail it cannot retain.

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses only the top two rows of the main map:[[1, 4, 2, 0], [3, 2, 5, 1]]. Its two complete windows give max[[4, 5]]and mean[[2.5, 2]], shape 1×2. Swapping1 and4 inside the left window changes positions but not its maximum 4or mean 2.5. That pair cannot be reconstructed from either unchanged scalar alone. The main 4×4 case has four coarse regions; do not copy its 2×2 output shape into this separate picture.

Trace unchanged summaries after a within-window swap



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Left window	Values 1, 4, 3, 2: max 4, mean 2.5.
2. Right window	Values 2, 0, 5, 1: max 5, mean 2.
3. Compare summaries	Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Change: Swap 1 and 4 within the left window.

Prediction: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Max pooling	Replacing each window by its largest value
Mean pooling	Replacing each window by the arithmetic mean of its values
Spatial information	Information about where values occur and how positions relate
Pooling collision	Different inputs producing the same pooled values

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Pooling and spatial information

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.

2. Right window

Values 2, 0, 5, 1: max 5, mean 2.

3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Teacher check: Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1.

A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Pooling and **spatial information**. Proposed explanation: "**Max pooling** turns negative numbers into zero". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: That is a ReLU behaviour, not the maximum of an all-negative window.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. The original 4 was top-right in the left window, but after swapping it is top-left; the pooled value 4 is identical. Changed-case check: Both pooled maps remain unchanged.

Max and mean ignore the positions of values inside the same window.

Changed-case reasoning: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate columns for window values, sum, divisor, maximum and mean. Use numbered positions rather than colour alone.

Extension: enumerate all sixteen binary 2 by 2 windows, group them by maximum and mean, and identify collisions under each summary.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve valid window starts and mark the four nonoverlapping regions.
7 to 18 minutes	Calculate both summaries in the first region, then complete W1.
18 to 32 minutes	Trace the remaining regions and run or hand-trace the optional code.
32 to 45 minutes	Use W2 and the visual to distinguish lost location from retained regional arrangement.
Session 2 start to 8 minutes	Complete the guided collision and discuss why storing a maximum position is still insufficient for reconstruction.
8 to 20 minutes	Use W3 to expose the omitted border, then trace the movement counterexample in W5.
20 to 35 minutes	Collect W4 independently, requiring signed sums and both maps without applying ReLU.
35 to 45 minutes	Use the exit response to qualify claims, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main complete sums, collisions and boundary probes supplied. W2 different4 positionsyieldmax4/mean 1. W3 incomplete99 borderexcluded, max 4/mean 2.5. W5 boundarymovementchangesmaps; do not infer unknown classifier outputs without its head. Independentwindow sums–10/12/16/–20, maxmap[[-1, 6], [7, –2]], mean[[-2.5, 3], [4, –5]]. Maxpoolingmustretainnegative maxima, not replace themwithzero.

Maxplusuniquepositioncanstilldiscardother values. No fitted weights or universal movement invariance is demonstrated.

Diagnose the current misconceptions

Max pooling turns negative numbers into zero: That is a ReLU behaviour, not the maximum of an all-negative window.

Pooling removes all **spatial information**: A local pooled grid retains coarse regional arrangement while losing within-window detail.

Saving the maximum position restores the **input**: Other values remain unknown even when a unique maximum location is recorded.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map.

Explain why replacing a negative maximum by zero is an error here.

Teacher explanation: Window sums in row order are -10, 12, 16 and -20. Max map is [[-1, 6], [7, -2]]. Mean map is [[-2.5, 3], [4, -5]].

Zero is not in either all-negative window; pooling is not ReLU and must return the largest actual value.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate both summaries

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

W2 Prove that location can be lost

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1. A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

W3 Audit an omitted border

Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

Teacher explanation: Shape is 1 by 1. Only [[1, 2], [3, 4]] is read, giving maximum 4 and mean 2.5. The final row and column lie outside the only complete stride -2 window.

W5 Check the movement boundary

A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

Teacher explanation: Before: max [[8, 0]], mean [[2, 0]]. After: max [[0, 8]], mean [[0, 2]]. Crossing a pooling boundary changes the **output**, refuting guaranteed invariance to every one-pixel movement.

▼ **Earlier wording, alternatives and picture transcripts**

Guided and exit questions

Guided: both windows have maximum 6 and mean 1.5. Either scalar alone cannot distinguish their different nonzero positions.

Exit: the first two visual windows collide, so exact reconstruction is impossible from the pooled values alone. The 8 crossing from column 1 to column 2 changes the pooled map, so universal shift invariance does not follow.

W1 Calculate both summaries

W2 Prove that location can be lost

W3 Audit an omitted border

W4 Pool a new map independently

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

Teacher explanation: Window sums in row order are -10, 12, 16 and -20. Max map is [[-1, 6], [7, -2]]. Mean map is [[-2.5, 3], [4, -5]].

Zero is not in either all-negative window; pooling is not ReLU and must return the largest actual value.

W5 Check the movement boundary

Independent assessment and fresh retry

W4 requires sums -10, 12, 16, -20, max map [[-1, 6], [7, -2]] and mean map [[-2.5, 3], [4, -5]]. Fresh retry: rows [-6, -2, 1, 5] and [-4, -8, 3, 7]. **Output** shape is 1 by 2.

Sums are -20 and 16; max map is [[-2, 7]] and mean map is [[-5, 4]]. Require a signed maximum rather than a zero-clamped answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate columns for window values, sum, divisor, maximum and mean. Use numbered positions rather than colour alone.

Extension: enumerate all sixteen binary 2 by 2 windows, group them by maximum and mean, and identify collisions under each summary.

Week 23 trains and evaluates a small image model with a declared baseline and data split.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/23, ai-5/25, ai-5/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Compute max and **mean pooling** under an explicit window rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Reasoned practice answer: Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1.

A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

Check every learner before discussion: Use the worked case for Pooling and **spatial information**. Proposed explanation: “**Max pooling** turns negative numbers into zero”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: That is a ReLU behaviour, not the maximum of an all-negative window.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Swap 1 and 4 within the left window. Require a **prediction** and a reason before comparing. Expected result: Both pooled maps remain unchanged. Reason: Max and mean ignore the positions of values inside the same window.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

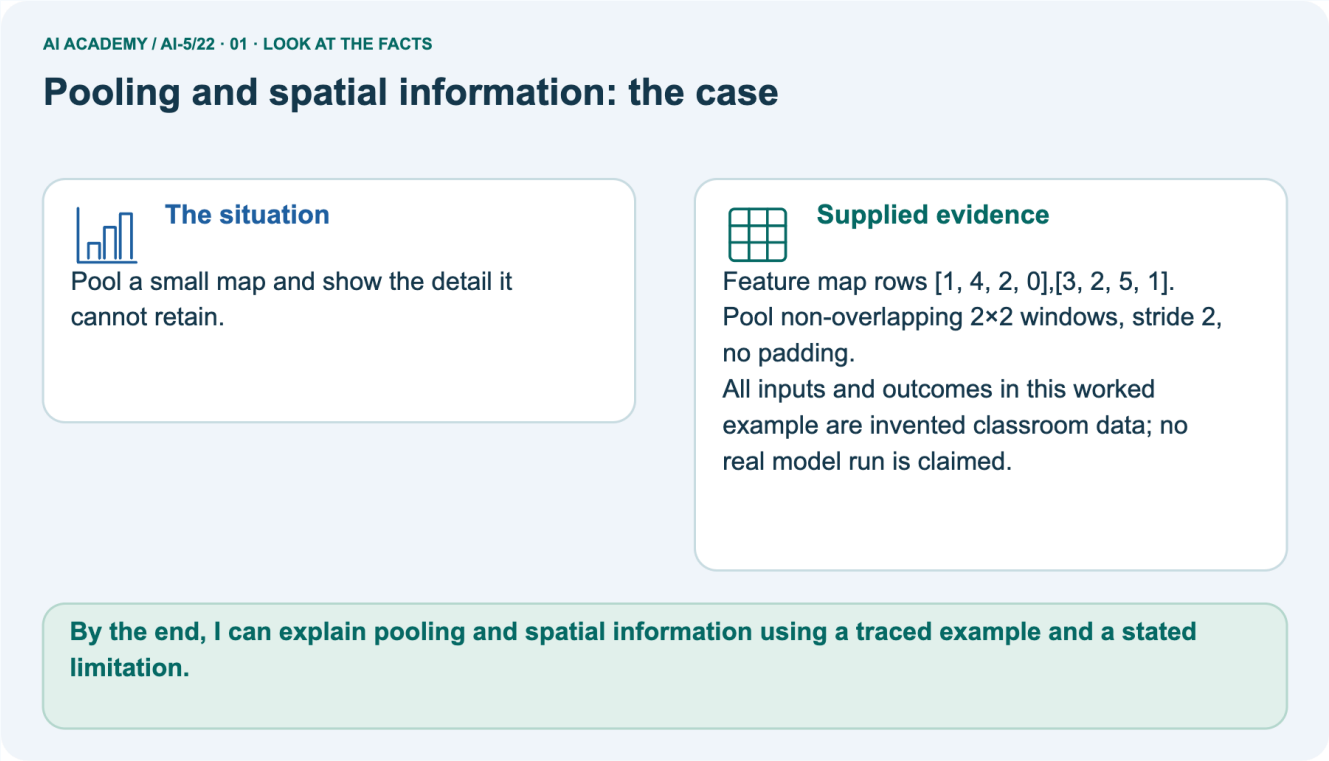
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Pooling and spatial information: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

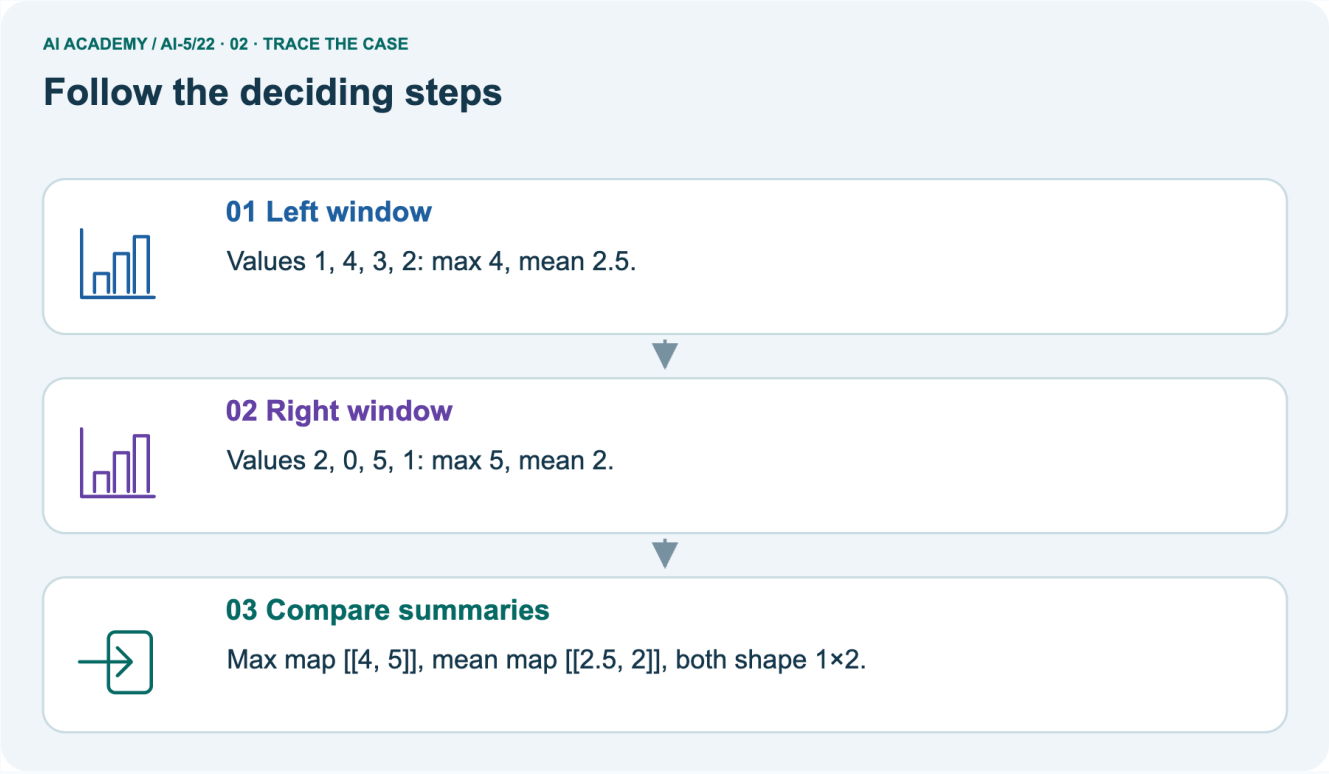
Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values 1, 4, 3, 2: max 4, mean 2.5. → Right window: Values 2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two adjacent pooling windows



Follow the labelled picture with the matching explanation. Open the vector diagram

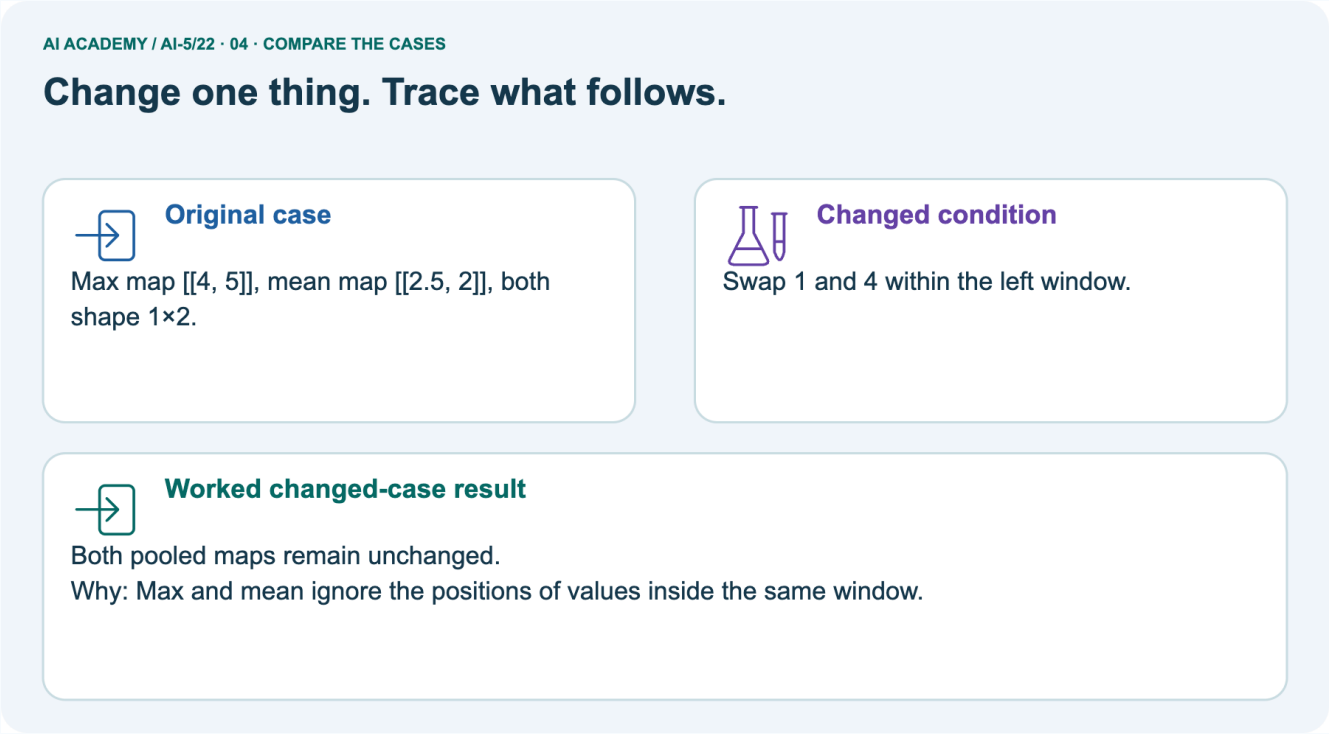
Two adjacent pooling windows. row 0: left 0 = 1, left 1 = 4, right 0 = 2, right 1 = 0. row 1: left 0 = 3, left 1 = 2, right 0 = 5, right 1 = 1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Changed condition: Swap 1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

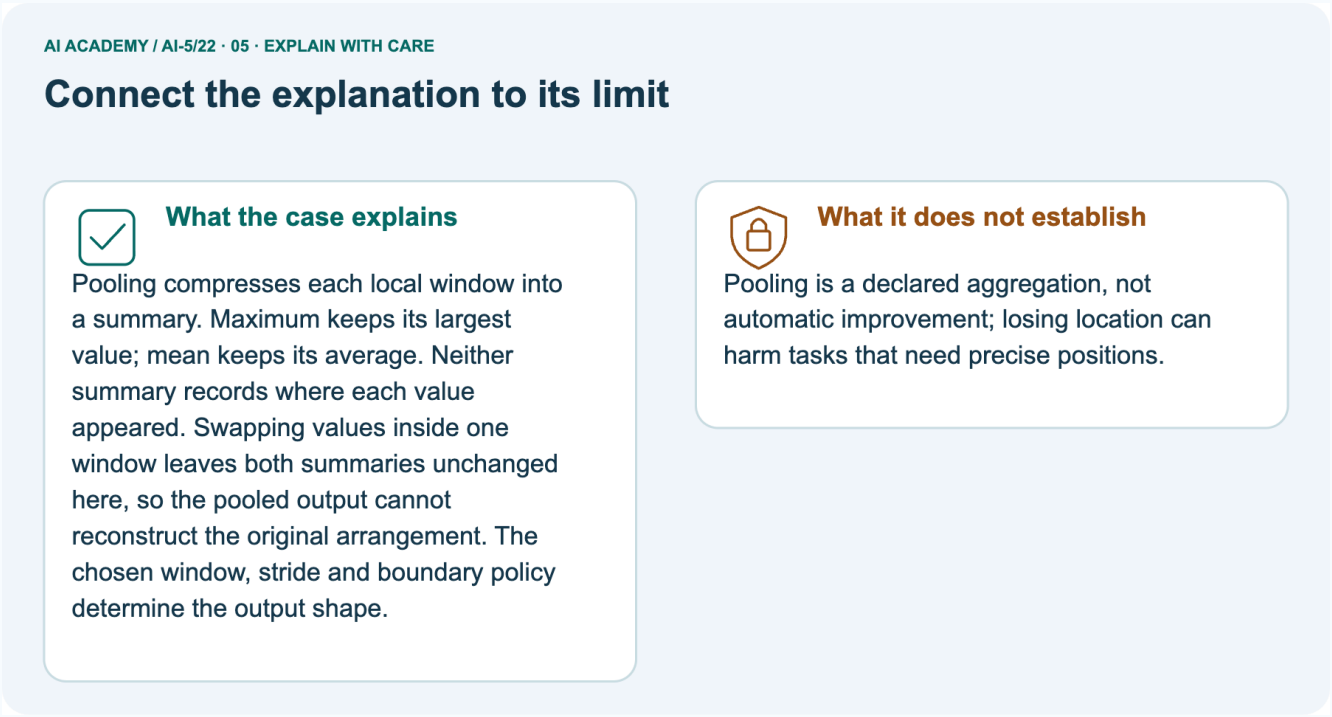
Reason: Max and mean ignore the positions of values inside the same window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Max and mean ignore the positions of values inside the same window.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

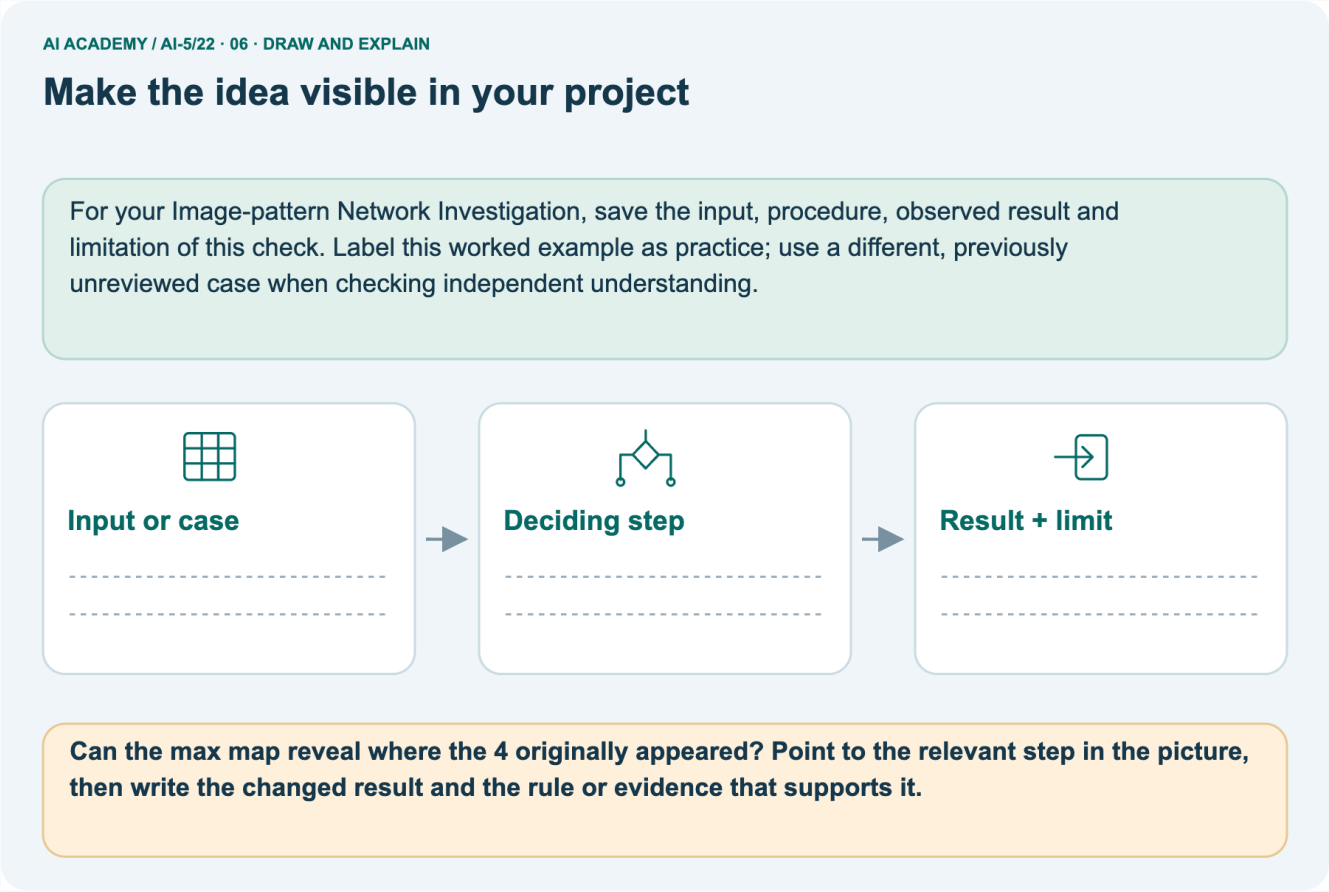
Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. The original 4 was top-right in the left window, but after swapping it is top-left; the pooled value 4 is identical. Changed-case check: Both pooled maps remain unchanged.

Max and mean ignore the positions of values inside the same window.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Pooling and spatial information: the case

02. Follow the deciding steps

03. Two adjacent pooling windows

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.